

ENVIRONMENT

AI TRIAGE SUPPORT SYSTEM CO-DESIGN CANVAS

What is the AI's context of operation?

Where

What place?
Does it change?

- Mercer Emergency Department triage desk.
- Shared clinical workstation used during patient assessment and reassessment.
- Busy, high-pressure environment with frequent interruptions and long waiting periods.
- Future expansion to other hospital departments and Observation Units.

User(s)

Who is using the robot?

- ED triage nurses (primary users).
- Nurses reviewing patient deterioration alerts submitted through the patient portal

When

What time of day?
Does it change?

- During every patient triage assessment.
- 24 hours per day.
- Peak demand periods and overnight shifts.
- During initial triage, reassessment and ongoing patient monitoring.

Secondary user(s)

Are there secondary users?
E.g. teachers that help students use a robot.

- Emergency physicians.
- Charge nurses.
- Bed managers.
- Clinical IT staff.
- Hospital governance staff.
- Observation Unit staff.

Data collection

Does the robot collect data from its environment?

- Patient demographics.
- Presenting complaint.
- Vital signs.
- Manual nurse assessment.
- Patient-reported symptom updates through the waiting portal.
- Future EHR integration.
- Prediction history, reassessment history and override logs.



TRADE-OFF:

More data collection requires more attention to data security.

Simultaneous users

How many users should be able to use the robot simultaneously?



TRADE-OFF:

More simultaneous users requires a more sophisticated robot.



Reason:

Multiple triage nurses may use the system simultaneously, but not dozens at one workstation.

External sensors and actuators

Does the robot use external sensors?
Does it have external actuators, such as lights or limbs?

- Manual or device-assisted vital-sign entry.
- Future integration with BP monitors, pulse oximeters and thermometers.
- No physical actuators.
- Visual alerts only.

Connection to systems

Is the robot connected to external systems, such as software, databases, or other robots?
How does it use these systems?

- Mercer Electronic Health Record.
- Hospital user authentication.
- Clinical audit database.
- Future reporting dashboard.
- Secure hospital network.
- Secure patient portal using QR-code authentication.